```
In[ ]:= Quit[];
```

# The Lorentzian quasi-Euclidean Kerr black hole (LEK)

```
In[ ]:= ClearAll["Global'*"];
        list[y_] := y[[1]] == y[[2]];
        rplc[y_] := y;
```

## Metric definition

We start by defining the coordinates vector using the Doran coordinate system {t, r, $\theta$, $\varphi$}

```
In[ ]:= x = {t, r, θ, φ}; n = 4;
```

and the function that implements the signature change $\epsilon$

```
In[ ]:= ϵ[r_] := ϵFunc[r];
```

We initialize the non-vanishing components of the LEK metric $g_{\mu\nu}$ in Doran coordinates

```
In[ ]:= g11 = gtt = -ϵ[r] (1 - (2 M r) / (r^2 + a^2 Cos[θ]^2));
        g22 = grr = (r^2 + a^2 Cos[θ]^2) / (r^2 + a^2);
        g33 = gθθ = r^2 + a^2 Cos[θ]^2;
        g44 = gφφ = (r^2 + a^2 + (2 M r a^2 Sin[θ]^2) / (r^2 + a^2 Cos[θ]^2)) Sin[θ]^2;
        g14 = gtφ = -Sqrt[ϵ[r]] (2 M r a Sin[θ]^2) / (r^2 + a^2 Cos[θ]^2);
        g12 = gtr = Sqrt[ϵ[r]] Sqrt[(2 M r) / (r^2 + a^2)];
        g24 = grφ = -Sqrt[((2 M r) / (r^2 + a^2))] a Sin[θ]^2;
        g23 = g34 = g13 = 0;
```

We now construct the metric tensor $g_{\mu\nu}$ and compute the inverse metric $g^{\mu\nu}$.
We also compute the determinant of the metric and verify the identity relation $g_{\mu\nu}\, g^{\mu\nu}$

```
In[ ]:= mt = FullSimplify[{{g11, g12, g13, g14},
          {g12, g22, g23, g24}, {g13, g23, g33, g34}, {g14, g24, g34, g44}}];
        it = FullSimplify[Inverse[mt]];
        mx = FullSimplify[
           Table[Simplify[Sum[it[[i, k]] * mt[[k, j]], {k, 1, n}]], {i, 1, n}, {j, 1, n}]];
        md = FullSimplify[Det[mt]];
        MatrixForm[mt]
        MatrixForm[it]
        MatrixForm[mx]
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} \left(-1+\frac{2\,M\,r}{r^2+a^2\,\text{Cos}[\theta]^2}\right)\in\text{Func}[r] & \sqrt{2}\,\sqrt{\frac{M\,r}{a^2+r^2}}\,\sqrt{\in\text{Func}[r]} & 0 & -\frac{2\,a\,M\,r\,\text{Sin}[\theta]^2\,\sqrt{\in\text{Func}}}{r^2+a^2\,\text{Cos}[\theta]^2} \\ \sqrt{2}\,\sqrt{\frac{M\,r}{a^2+r^2}}\,\sqrt{\in\text{Func}[r]} & \frac{r^2+a^2\,\text{Cos}[\theta]^2}{a^2+r^2} & 0 & -\sqrt{2}\,a\,\sqrt{\frac{M\,r}{a^2+r^2}}\,\text{Sin}[ \\ 0 & 0 & r^2+a^2\,\text{Cos}[\theta]^2 & 0 \\ -\frac{2\,a\,M\,r\,\text{Sin}[\theta]^2\,\sqrt{\in\text{Func}[r]}}{r^2+a^2\,\text{Cos}[\theta]^2} & -\sqrt{2}\,a\,\sqrt{\frac{M\,r}{a^2+r^2}}\,\text{Sin}[\theta]^2 & 0 & \text{Sin}[\theta]^2\left(a^2+r^2+\frac{2\,a^2\,M\,r}{r^2+a^2}\right. \end{pmatrix}$$

Out[ ]//MatrixForm=

$$\begin{pmatrix} -\frac{1}{\in\text{Func}[r]} & \frac{\sqrt{2}\,M\,r}{\sqrt{\frac{M\,r}{a^2+r^2}}\,\left(r^2+a^2\,\text{Cos}[\theta]^2\right)\,\sqrt{\in\text{Func}[r]}} & 0 & 0 \\ \frac{\sqrt{2}\,M\,r}{\sqrt{\frac{M\,r}{a^2+r^2}}\,\left(r^2+a^2\,\text{Cos}[\theta]^2\right)\,\sqrt{\in\text{Func}[r]}} & \frac{2\left(a^2+r\,(-2\,M+r)\right)}{a^2+2\,r^2+a^2\,\text{Cos}[2\,\theta]} & 0 & \frac{\sqrt{2}\,a\,\sqrt{\frac{M\,r}{a^2+r^2}}}{r^2+a^2\,\text{Cos}[\theta]^2} \\ 0 & 0 & \frac{1}{r^2+a^2\,\text{Cos}[\theta]^2} & 0 \\ 0 & \frac{\sqrt{2}\,a\,\sqrt{\frac{M\,r}{a^2+r^2}}}{r^2+a^2\,\text{Cos}[\theta]^2} & 0 & \frac{\text{Csc}[\theta]^2}{a^2+r^2} \end{pmatrix}$$

Out[ ]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

# Curvature related quantities including explicit form of Ricci and regularized Kretschmann scalar

Christoffel symbols $\Gamma^{\alpha}{}_{\mu\nu}$

```
In[ ]:= chr = ParallelTable[FullSimplify[
            (1 / 2) Sum[(it[[i, s]]) (D[mt[[s, j]], x[[k]]] + D[mt[[s, k]], x[[j]]] - D[mt[[j, k]], x[[s]]]),
              {s, 1, n}]], {i, 1, n}, {j, 1, n}, {k, 1, n}];
```

Riemann curvature tensor $R^{\beta}{}_{\mu\nu\alpha}$

```
In[ ]:= rmn = ParallelTable[FullSimplify[D[chr[[i, j, l]], x[[k]]] - D[chr[[i, j, k]], x[[l]]] +
              Sum[chr[[s, j, l]] * chr[[i, k, s]] - chr[[s, j, k]] * chr[[i, l, s]], {s, 1, n}]],
           {i, 1, n}, {j, 1, n}, {k, 1, n}, {l, 1, n}];
```

Ricci tensor $R_{\mu\nu}$

```
In[ ]:= rcc = ParallelTable[
           FullSimplify[Sum[rmn[[i, j, i, l]], {i, 1, n}]], {j, 1, n}, {l, 1, n}];
```

Ricci scalar $R$

```
In[•]:= R = FullSimplify[Sum[it[[i, j]] × rcc[[i, j]], {i, 1, n}, {j, 1, n}]]
```

Out[•]=

$$\frac{1}{\left(a^2+2\,r^2+a^2\,\text{Cos}[2\,\theta]\right)^3\,\epsilon\text{Func}[r]^2}\left(4\,M\,r^2\left(a^2+r^2\right)\epsilon\text{Func}'[r]\left(-4\,\epsilon\text{Func}[r]+M\,\epsilon\text{Func}'[r]\right)+2\,M\left(a^2+2\,r^2+a^2\,\text{Cos}[2\,\theta]\right)\epsilon\text{Func}'[r]\left(2\left(a^2+r^2\right)\epsilon\text{Func}[r]-M\,r^2\,\epsilon\text{Func}'[r]\right)+\left(a^2+2\,r^2+a^2\,\text{Cos}[2\,\theta]\right)^2\left(\epsilon\text{Func}'[r]\left(4\,(M-r)\,\epsilon\text{Func}[r]+\left(a^2+r\,(-2\,M+r)\right)\epsilon\text{Func}'[r]\right)-2\left(a^2+r\,(-2\,M+r)\right)\epsilon\text{Func}[r]\,\epsilon\text{Func}''[r]\right)\right)$$

When sending epsilon derivatives to zero we recover the standard result for a Kerr black hole (R=0)

```
In[•]:= R /. {D[ϵ[r], r] → 0, D[D[ϵ[r], r] → 0, r]}
```

Out[•]=

```
0
```

Fully covariant Riemann tensor $R_{\mu\nu\alpha\beta}$

```
In[•]:= rcv = ParallelTable[FullSimplify[Sum[mt[[i, j]] * rmn[[j, k, l, m]], {j, 1, n}]],
    {i, 1, n}, {k, 1, n}, {l, 1, n}, {m, 1, n}];
```

Fully contravariant Riemann tensor $R^{\mu\nu\alpha\beta}$

```
In[•]:= rcn = ParallelTable[FullSimplify[
     Sum[it[[m, i]] * it[[h, j]] * it[[o, k]] * it[[p, l]] * rcv[[i, j, k, l]], {i, 1, n},
      {j, 1, n}, {k, 1, n}, {l, 1, n}]], {m, 1, n}, {h, 1, n}, {o, 1, n}, {p, 1, n}];
```

Kretschmann scalar $K = R_{\mu\nu\alpha\beta}R^{\mu\nu\alpha\beta}$ and regularized form with derivatives of $\epsilon$ nullified

```
In[•]:= krn = FullSimplify[
    Sum[rcv[[i, j, k, l]] * rcn[[i, j, k, l]], {i, 1, n}, {j, 1, n}, {k, 1, n}, {l, 1, n}]];
```

```
In[•]:= regkrn = krn /. {D[ϵ[r], r] → 0, D[D[ϵ[r], r], r] → 0} // FullSimplify
```

Out[•]=

$$-\left(\left(192\,M^2\left(a^2-2\,r^2+a^2\,\text{Cos}[2\,\theta]\right)\left(3\,a^4-56\,a^2\,r^2+8\,r^4+4\,a^2\left(a^2-14\,r^2\right)\text{Cos}[2\,\theta]+a^4\,\text{Cos}[4\,\theta]\right)\right)\Big/\left(a^2+2\,r^2+a^2\,\text{Cos}[2\,\theta]\right)^6\right)$$

Fully covariant Weyl tensor $C_{\mu\nu\alpha\beta}$

```
In[•]:= wcv = ParallelTable[
    FullSimplify[rcv[[i, j, k, l]] - 1/(n - 2) (mt[[i, k]] × rcc[[j, l]] - mt[[i, l]] × rcc[[j, k]] -
        mt[[j, k]] × rcc[[i, l]] + mt[[j, l]] × rcc[[i, k]]) +
      R/((n - 1) (n - 2)) (mt[[i, k]] × mt[[j, l]] - mt[[i, l]] × mt[[j, k]])],
    {i, 1, n}, {j, 1, n}, {k, 1, n}, {l, 1, n}];
```

Fully contravariant Weyl Tensor $C^{\mu\nu\alpha\beta}$

```
In[•]:= wcn = ParallelTable[FullSimplify[
     Sum[it[[m, i]] * it[[h, j]] * it[[o, k]] * it[[p, l]] * wcv[[i, j, k, l]], {i, 1, n},
      {j, 1, n}, {k, 1, n}, {l, 1, n}]], {m, 1, n}, {h, 1, n}, {o, 1, n}, {p, 1, n}];
```

Weyl tensor $C^{\mu}{}_{\nu\alpha\beta}$

In[•]:=
```
weyl = ParallelTable[FullSimplify[Sum[it[[i, s]] × wcv[[s, j, k, l]], {s, 1, n}]],
    {i, 1, n}, {j, 1, n}, {k, 1, n}, {l, 1, n}];
```

Weyl tensor norm C

In[•]:=
```
cnorm = FullSimplify[
    Sum[wcv[[i, j, k, l]] * wcn[[i, j, k, l]], {i, 1, n}, {j, 1, n}, {k, 1, n}, {l, 1, n}]];
```

Final verification that Weyl tensor norm should be equal to Kretschmann scalar

In[•]:=
```
(cnorm - krn) /. {D[ϵ[r], r] → 0, D[D[ϵ[r], r], r] → 0} // FullSimplify
```

Out[•]=
```
0
```